



A Deep Dive into the US Electric Vehicle Industry

Driving the Electric Future

Suhrid Barua

US Electric Vehicle Industry Whitepaper

Introduction

The U.S. electrical vehicle industry is much older than commonly perceived. EVs first emerged in the late 1890s when the American automotive landscape was still unsettled and competing technologies - gasoline, steam, and electric propulsion competed for adoption on American roads. Electric vehicles in that era, were considered practical and desirable, offering quiet, clean, and effortless driving experience, considered ideal for urban environments. By 1900, EVs represented nearly one-third of all vehicles in the United States, with strong adoption among affluent urban households in cities such as New York, Boston, and Chicago.

Several pioneering companies such as Columbia Automobile Company, Baker Motor Vehicle Company, and Detroit Electric shaped the early era of electric mobility. However, by the 1920s, electric vehicles lost ground to mass-produced internal combustion engines as range limitations, high battery costs, and lack of charging infrastructure constrained scalability. The introduction of affordable gasoline cars, led by the Ford Model T, pushed electric vehicles out of the mainstream for many years, limiting them to niche applications such as industrial equipment and specialized fleets.



Tesla Y Model is the highest-selling EV in the US

EVs witnessed a meaningful resurgence in the late 1990s and early 2000s. Rising environmental awareness, stricter emissions standards, and advances in battery and software technology reignited automaker interest. General Motors (GM) launched the EV1 in 1996, followed by Nissan's Altra EV in 1998. While these vehicles remained niche with limited range, they reintroduced electrification into public consciousness and laid the groundwork for broader acceptance.

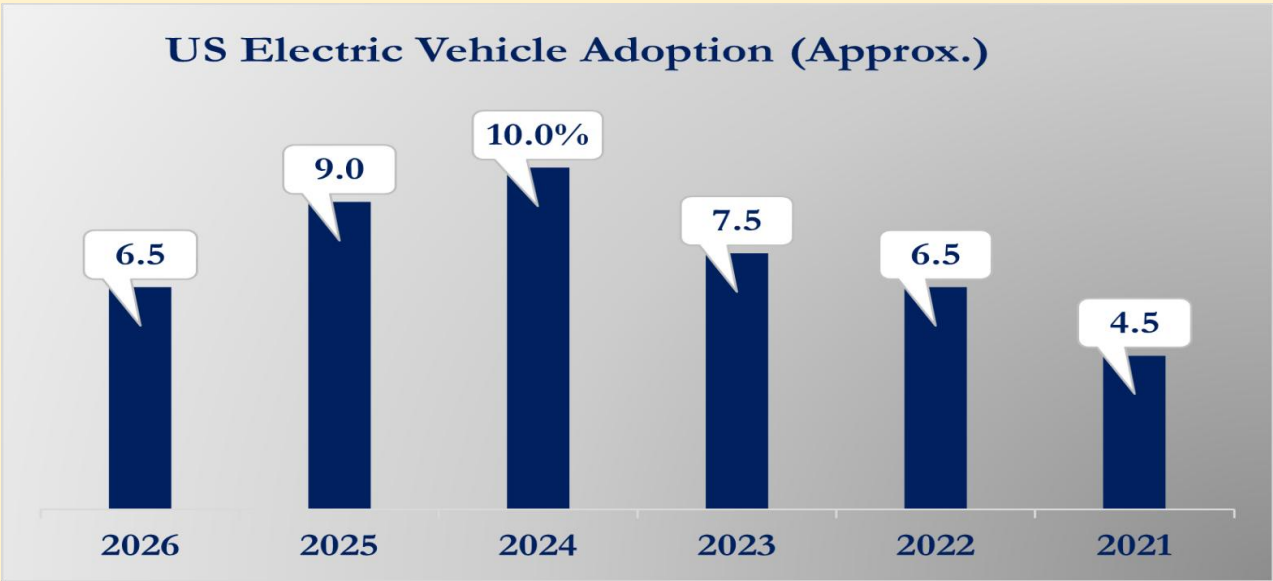
Tesla's entry in the EV space with the Roadster in 2008 did not start this wave, but it fundamentally redefined it. It demonstrated that battery-electric vehicles (BEVs) could be aspirational, high-performance, and commercially viable, setting new benchmarks for range, software integration, and user experience.

From early experimentation to today’s large-scale commercialization, EVs have evolved into a core automotive segment. This evolution has also established high expectations around performance, infrastructure, and scalability - standards that continue to shape the highly competitive U.S. EV market today.

US EV Market Overview

The EV market in the US has evolved from an early-adoption phase into a mid-growth expansion stage, largely driven by federal incentives, tightening emissions regulations, and rapid product diversification across legacy automakers and EV startups. EV sales rose from approximately 450,000 units in 2021 to 1.3 million units in 2024, according to Cox Automotive’s Kelley Blue Book, reflecting a strong multi-year adoption curve. However, the EV momentum appears to have moderated in 2025 and early 2026, with Q1 2026 sales of around 216,000 units, around a 27% year-on-year decline suggesting a short-term slowdown in demand, bringing volumes closer to late-2022 levels.

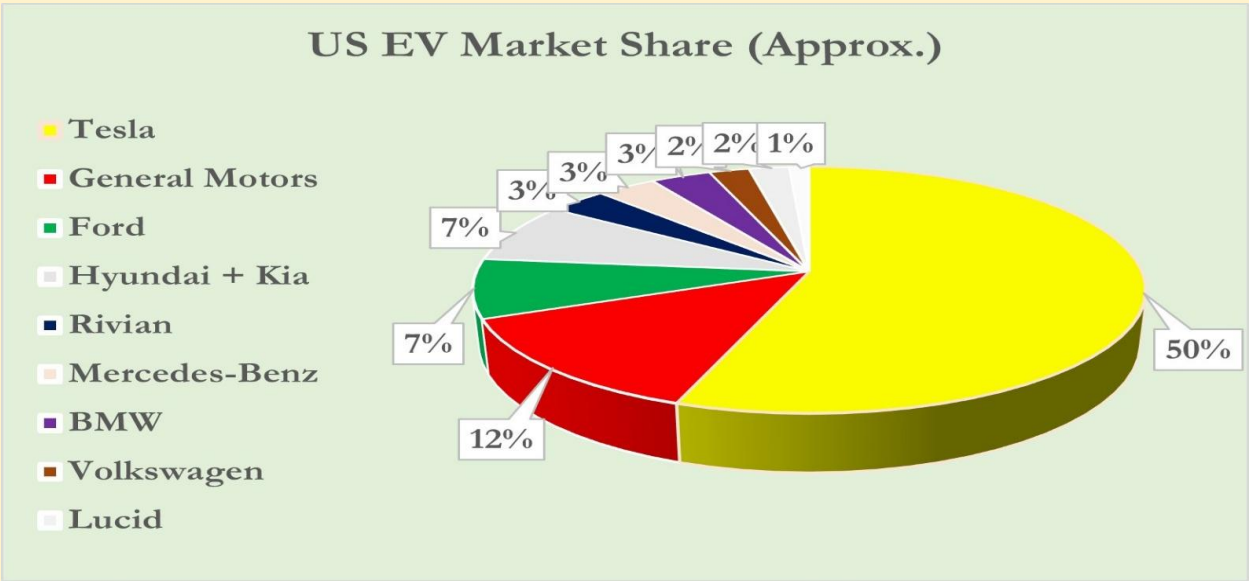
This moderation does not reflect weakening demand, but rather a structural reset in market dynamics. While global EV sales expanded over 20% in 2025 to exceed 20 million units, with roughly one in four new vehicles sold globally now electric as per International Energy Agency (IEA) estimates, the U.S. has diverged from global momentum. U.S. EV sales declined modestly in 2025 despite global acceleration, highlighting a growing gap between domestic policy structure and international adoption dynamics.



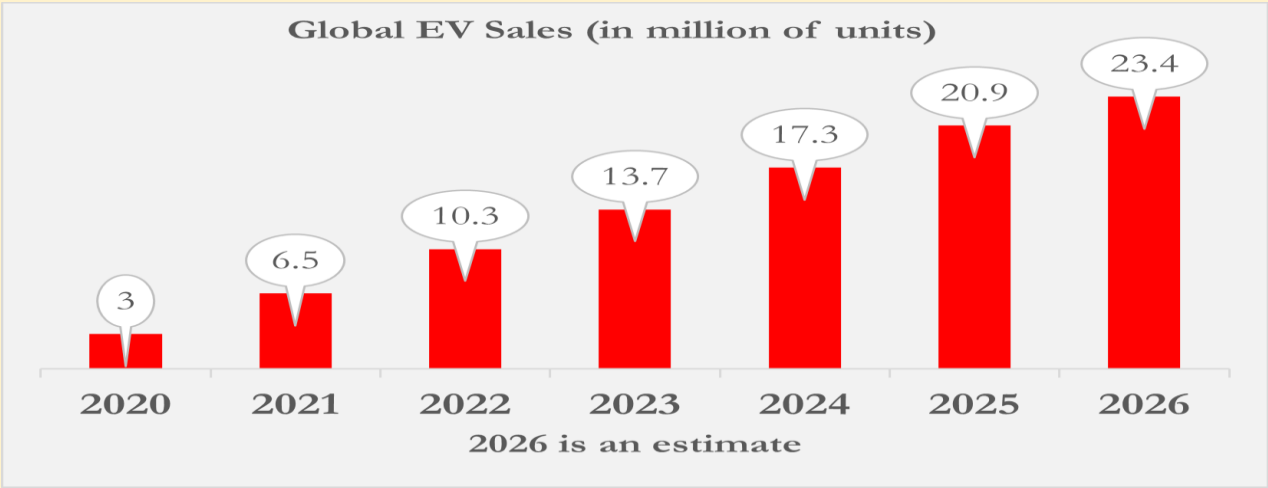
The divergence can be attributed more to supply-side and policy constraints than to weak consumer demand. U.S. buyers have limited access to lower-cost EV imports, particularly from China, due to high tariffs and regulatory restrictions. In addition, the expiration of the \$7,500 federal EV tax credit in 2025 has reduced affordability at the entry-level segment. Together, these factors have constrained price accessibility and slowed mass-market adoption, even as fuel price dynamics and global electrification trends continue to support EV demand elsewhere. In contrast, Europe, Asia-Pacific, and Latin America recorded significantly stronger EV growth in 2025, with Europe up ~30%, Asia-Pacific (excluding China) up ~80%, and Latin America up ~75%, according to International Energy Agency (IEA) estimates.

Regionally, EVs are gaining faster adoption across Europe. Norway leads global adoption with ~98% penetration, followed by Denmark at ~80%. Major markets such as the UK (~26%), Germany (~23%), and France (~23%)

reflect a steady shift to EVs. India continues to expand its EV base and is expected to reach ~8–10% penetration by 2026-end.



Overall, the U.S. EV market remains a high-value but structurally constrained transition market, shifting from incentive-led expansion toward fundamentals-driven adoption. While long-term growth is supported is supported by improving battery costs, expanding charging infrastructure, and broader model availability, the near-term outlook reflects a shift from rapid growth to a more normalized and uneven adoption phase within a fragmented global EV landscape.



Market Leaders

- Tesla
- GM
- Ford

Legacy OEM Challengers

- Hyundai/Kia
- BMW
- Mercedes-Benz
- Volkswagen

EV Pure-Play Startups

- Rivian
- Lucid

Commercial & Specialty OEMs

Transit Bus: New Flyer, BYD, Gillig, EDC, Phoenix Motor.

School Bus: Blue Bird, Thomas Built Buses, BYD, IC Bus.

Pickup Truck & Last-Mile Delivery Vehicles (Electric Delivery Vans & Urban Logistics)

- Legacy OEM EVs: GM, Ford
- EV Pure-Play: Rivian

Medium & Heavy-Duty Truck: Daimler Truck North America, Volvo Trucks North America, BYD, PACCAR, International Motors (formerly Navistar), Hino Motors, Motif Power Systems, Orange EV, Xos, Einride, Harbinger, Windrose Technology.

Electric Scooter OEMs: Razor, Segway-Ninebot, NIU Technologies.

Electric Shared Mobility Scooter OEMs: Lime (owned by Neutron Holdings, Inc.), Bird & Spin (owned by Third Lane Mobility).

Electric Motorcycle OEMs: Zero Motorcycles, Lightning Motorcycles, LiveWire (Earlier part of Harley-Davidson).

Electric Three-Wheeler OEMs: Arcimoto, Aptera Motors.

Competitive Landscape

The U.S. electric vehicle (EV) market is increasingly defined by competition between Tesla, legacy automakers, and emerging startups, where execution speed, scale, and software-led differentiation determine positioning. However, this domestic structure is increasingly influenced by global dynamics, especially China's dominance in affordable EV manufacturing and a widening global cost gap in EV supply.

Tesla continues to dominate the U.S. EV market, with around 46% share at the end of 2025, down from roughly 49% in 2024. Its vertically integrated manufacturing model, software-defined vehicle ecosystem, and control over

battery strategy, charging infrastructure, and over-the-air software capabilities have enabled sustained cost leadership, faster product iteration, and stronger profitability compared to most competitors, reinforcing Tesla's position as the benchmark for scale and efficiency in the U.S. EV market.

Legacy OEMs - Ford, General Motors, and Toyota - are scaling electrification programs but remain constrained by legacy ICE infrastructure, slower development cycles, and high capital intensity. While platforms such as GM's Ultium architecture and Ford's restructuring efforts have improved their competitiveness, most legacy automakers still lag behind Tesla, Inc. in software integration, charging ecosystem development, and cost efficiency. As a result, incumbents remain in a transitional phase, scaling up their EV portfolios while still working through profitability challenges.



Tesla Cybertruck Electric Pickup Truck

Emerging EV startups such as Rivian and Lucid represent innovation-led challengers, while others highlight execution and funding risks in scaling production. Across this segment, access to capital, manufacturing efficiency, and supply chain resilience remain key survival factors. Beyond domestic players, global competition, led by China and supported by Europe and parts of Asia, continues to intensify, driven by cost-competitive EV platforms and aggressive export strategies, reshaping the long-term competitive landscape for the U.S. market.

EV OEM Bankruptcies: A Drag on the U.S. EV Market

The U.S. electric vehicle transition underpins a consistent reality: EV adoption has not been limited by demand, but by the industry's inability to convert ambition into scalable, reliable, and profitable manufacturing systems. A wave of OEM bankruptcies has exposed recurring weaknesses in execution discipline, capital efficiency, and fleet-grade product readiness.

Several EV startups across segments have faced financial distress or bankruptcy. Truck OEMs Nikola Corporation and Lion Electric filed for Chapter 11 bankruptcy, while light commercial vehicle players such as Lordstown Motors, Canoo, and Electric Last Mile Solutions followed similar paths. Proterra an early leader in U.S. electric buses, achieved initial commercial success before facing financial distress and being acquired by Phoenix Motor Inc. In

shared mobility, scooter operators Bird and Spin were acquired by Third Lane Mobility after scaling challenges. Electric motorcycle startup Fuel Inc. and EV manufacturer Fisker Inc. also faced insolvency pressures, highlighting broader sector-wide consolidation.

EV Supply Chain, Battery Chemistry & Charging Infrastructure

The U.S. electric vehicle ecosystem is fundamentally shaped by its battery supply chain, where cell manufacturing and raw material dependencies remain the single largest cost and scalability driver, constituting around 30–40% of total EV cost. LG Energy Solution and SK On collectively form the backbone of domestic battery production. LG Energy Solution has emerged as a key U.S. supplier across multiple OEM programs, while SK On plays a central role in Ford's EV lineup. CATL, a global leader in battery production, has a limited U.S. presence.

Battery chemistry in the U.S. is increasingly segmented by vehicle class and use case. Lithium Iron Phosphate (LFP) batteries are gaining strong traction in entry-level passenger hatchbacks, sedans, and cost-sensitive fleet applications due to their lower cost, higher thermal stability, and longer lifecycle. In contrast, Nickel Manganese Cobalt (NMC) batteries remain dominant in premium sedans, SUVs, crossovers, and electric pickup trucks, where higher energy density and extended range are critical performance requirements. In electric bus and truck segments, both chemistries are used. Having said that, LFP adoption is rising rapidly due to its durability, safety profile, and total cost advantages in high-utilization fleet operations.

Charging infrastructure in the U.S. continues to expand but remains uneven across geography and use cases. It is led by Tesla Superchargers, Electrify America, and a growing ecosystem of OEM-backed charging networks, with the industry gradually converging toward the North American Charging Standard (NACS), improving interoperability and reducing fragmentation. However, there are still gaps in charging and EV access in rural areas, dense city housing, and long-distance trucking routes, leading to uneven adoption across states. As a result, the U.S. EV ecosystem reflects a dual reality: it is growing fast in local manufacturing and battery partnerships but still depends on global supply chains while continuing to build out its charging network.



Rivian R1T Electric Pickup Truck

Overview of US Electric Vehicle Segments

(A) Electric Passenger Car Segment

The U.S. electric passenger car market spans hatchbacks, sedans, SUVs, crossovers, and luxury vehicles, reflecting diverse adoption patterns shaped by consumer preference, vehicle utility, and pricing dynamics. Over time, demand has shifted from smaller cars to larger, more practical vehicle types. Electric hatchbacks remain a small segment, with early models like the Nissan Leaf and Chevrolet Bolt EV helping initial adoption. Other small urban EVs such as the BMW i3, Mini Cooper SE, and Fiat 500e contributed to early awareness but have seen limited long-term growth as U.S. buyers increasingly prefer larger, more spacious vehicles.

SUVs and crossovers now form the dominant segment of U.S. EV adoption, led by Tesla, Inc. Model Y, which sets benchmarks for scale, range, and software integration. Legacy OEMs such as Ford, General Motors, and Volkswagen have also expanded strongly with models like the Mustang Mach-E, Equinox EV, Lyriq, and ID.4. This segment benefits from strong consumer preference for larger vehicles, improving charging infrastructure, and a widening product portfolio.

Luxury EVs remain a smaller but important segment, focusing on performance, long range, and advanced technology. Overall, the market has shifted from hatchbacks enabling early adoption to SUVs driving mass growth, while luxury EVs continue to showcase innovation and performance leadership.



Daimler North America's Freightliner electric truck

(B) Electric Pickup Truck & Last-Mile Delivery/MPV Segment

The pickup truck segment is the backbone of the U.S. EV market. The segment is dominated by General Motors, Ford Motor Company alongside EV startup Rivian and Tesla. GM offers multiple models in this space, including the Chevrolet Silverado EV, GMC Hummer EV, and GMC Sierra EV. The Ford F-150 Lightning – one of the top-selling electric pickup trucks in the US - has been discontinued as Ford rejigs its focus on more compact and affordable vehicles. The segment is also expected to see new entrants, including Slate Auto's no-frills pickup backed by Amazon founder Jeff Bezos, and Scout Motors' upcoming Scout Terra under Volkswagen Group. Both Slate and Terra are expected to arrive in the US market by 2026-end or early 2027.

The last-mile delivery and MPV segment is one of the fastest-growing segments, driven by fleet electrification across logistics networks. OEMs such as Rivian, Ford, and GM's BrightDrop Zevo vans are increasingly being deployed by large fleet operators including Amazon, FedEx, and Walmart. However, scaling remains dependent on charging infrastructure availability and depot-level deployment readiness.

(C) Electric Bus Segment

The U.S. electric bus market is expanding rapidly, driven by zero-emission mandates, fleet replacement cycles, and public funding. It includes both transit buses for urban transport and school buses for student mobility, with both segments moving from pilot deployments toward broader fleet-level adoption across states. Proterra was an early mover in this space around 2010, achieving initial commercial success (it was acquired by Phoenix Motor in 2023), followed by the entry and expansion of players such as BYD, New Flyer, and Gillig.

The electric school bus segment gained momentum around 2018, supported by federal and state incentives, and has been led by Blue Bird, Thomas Built Buses, and IC Bus. Overall, transit buses now represent a more mature and consolidated segment, while school buses remain in a high-growth phase driven by continued policy support and fleet electrification programs.

(D) Electric Medium & Heavy-Duty Truck Segment

The U.S. electric truck segment started gaining momentum around 2018, led by BYD Motors, followed by established OEMs such as Daimler Truck North America, Volvo North America, PACCAR, Hino Motors, and International Motors, alongside specialized players like Motif Power Systems and Orange EV, and startups including Xos, Einride, Harbinger, and Windrose Technology. Early adoption was concentrated in ports, urban delivery routes, and niche vocational applications before expanding toward broader fleet electrification.

Daimler Truck North America has been a key player through its Freightliner brand, while Volvo North America and PACCAR have expanded offerings via models such as the Volvo VNR Electric, Kenworth, and Peterbilt electric trucks. Overall, medium-duty trucks are gaining traction, while heavy-duty trucks are still in an early scaling phase focused on regional freight and fleet adoption.



New Flyer's electric transit bus – one of the leading electric bus OEMs in the US

(E) Electric Two-Wheeler Segment (Scooters + Motorcycles)

The U.S. electric two-wheeler market has developed along two parallel tracks: scooters for urban mobility and shared micro-mobility, and motorcycles for personal transport and performance use. Scooters initially gained traction through early consumer products and later expanded rapidly via shared mobility platforms, before consolidating into fewer large operators as unit economics and regulatory pressures reshaped the market.

Razor USA, Segway-Ninebot, and NIU Technologies helped establish the consumer and connected scooter segment, while shared mobility players such as Lime and Spin drove large-scale urban adoption. In motorcycles, Zero Motorcycles has led the category since 2009, with LiveWire representing legacy OEM entry into electrification. Overall, the segment is gradually maturing, shifting from early experimentation to more structured, technology-driven urban mobility adoption.

(F) Electric Three-Wheeler Segment

The U.S. electric three-wheeler segment remains a niche but emerging mobility category positioned between two-wheelers and compact EVs. It primarily serves short-range urban commuting and light last-mile delivery use cases, offering high maneuverability, low operating cost, and efficiency in dense city environments. Growth is still early-stage, with demand driven by micro-mobility needs and urban sustainability trends.

Arcimoto is currently the only U.S. OEM producing and delivering electric three-wheelers at limited commercial scale through its FUV (Fun Utility Vehicle) platform. Aptera Motors remains in pre-production and validation phase, progressing toward initial manufacturing and customer deliveries. Overall, the segment is still developing, with future growth dependent on scaling capability, regulatory alignment, and supporting urban mobility infrastructure.

U.S. EV Policy, Incentives & Regulatory Framework

The U.S. EV industry has been shaped by federal and state policies that have evolved since EV incentives began in 2008. Early government support came through consumer tax credits, which helped boost EV adoption and benefited companies like Tesla, Inc. and General Motors in the early market stage.

A major shift came with the Inflation Reduction Act (IRA) of 2022, which moved EV incentives from broad consumer subsidies to support for domestic supply chains. The \$7,500 tax credit was redesigned with requirements like North American assembly and sourcing of critical minerals and battery components. It also extended support to used EVs and commercial vehicles, making incentives more targeted rather than universal.

However, the policy environment has since tightened, with federal EV tax credits effectively withdrawn in September 2025. This has reduced purchase incentives and made entry-level EVs less affordable, leading to slower or softer demand as buyers adjust to higher prices without subsidies. State incentives and federal charging programs still support the market, but the overall framework now focuses more on balancing demand with domestic manufacturing and supply-chain development.

Future Outlook

The U.S. EV industry is shifting from policy-driven growth to a phase focused on cost, scale, and ecosystem strength, with EVs expected to reach 25–30% of new vehicle sales in the next decade, though adoption will vary across segments and income groups. The withdrawal of federal EV tax credits in 2025 may soften near-term demand, especially in entry- and mid-market segments, as higher prices make buyers more cautious. Despite this, the long-term outlook remains strong, with Tesla, Inc. expected to maintain leadership while legacy OEMs focus on cutting costs and improving batteries and platforms. Overall, the market is moving toward fundamentals-led competition, where affordability, technology integration, and charging ecosystem strength will determine long-term winners.